

Blockchain Technology: Principles and Applications

A blockchain is a distributed ledger that records transactions across many computers so that the record cannot be altered retroactively without alteration of all subsequent blocks and the consensus of the network. Bitcoin, introduced in 2009 by the pseudonymous Satoshi Nakamoto, was the first practical implementation.

1. Core Concepts

Each block contains a cryptographic hash of the previous block, a timestamp, and transaction data. The chain is immutable: changing one block invalidates every subsequent block. Consensus mechanisms—Proof of Work (PoW) and Proof of Stake (PoS)—ensure agreement among distributed nodes without a central authority.

2. Smart Contracts

Smart contracts are self-executing programs stored on a blockchain that automatically enforce the terms of an agreement when predefined conditions are met. Ethereum, launched in 2015, popularised smart contracts and underpins most DeFi (decentralised finance) and NFT (non-fungible token) applications.

3. Decentralised Finance (DeFi)

DeFi replicates traditional financial services—lending, borrowing, trading—using smart contracts, eliminating intermediaries such as banks. Automated market makers (AMMs) like Uniswap allow token swaps without order books. Total value locked (TVL) in DeFi protocols peaked at over 180 billion USD in late 2021.

4. Enterprise Blockchain

Permissioned blockchains such as Hyperledger Fabric and R3 Corda are used in supply chain management, trade finance, and healthcare to improve transparency and reduce reconciliation costs. IBM Food Trust uses Hyperledger to trace produce from farm to shelf.

5. Challenges and Outlook

Scalability (the blockchain trilemma), energy consumption (PoW), regulatory uncertainty, and user experience remain key challenges. Layer-2 solutions like the Lightning Network and Ethereum rollups address throughput. Central bank digital currencies (CBDCs) are being explored by over 100 countries as government-backed digital money.